EXPERIMENT – 5

Perform various configurations of IPTABLES Firewall in Linux

AIM:

To study and perform various configurations of the IPTABLES firewall in Linux to control incoming and outgoing network traffic.

REQUIREMENTS:

1. Linux Operating System (Ubuntu / Kali Linux)

2. Root or sudo privileges

3. IPTABLES firewall utility

PROCEDURE:

1. Open the terminal in Linux.

2. Check the current IPTABLES rules using iptables -L.

3. Flush existing rules using iptables -F.

4. Set default policies for INPUT, OUTPUT, and FORWARD chains.

5. Allow SSH traffic.

6. Block ICMP (ping) requests.

7. Allow HTTP and HTTPS traffic.

8. Save the firewall rules.

CONFIGURATIONS / SOURCE CODE:

```
# View current rules

iptables -L
```

```
# Flush all existing rules
iptables -F

# Set default policies
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

# Allow localhost traffic
iptables -A INPUT -i lo -j ACCEPT

# Allow established connections
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

# Allow SSH (port 22)
iptables -A INPUT -p tcp --dport 22 -j ACCEPT

# Allow HTTP and HTTPS
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
iptables -A INPUT -p tcp --dport 443 -j ACCEPT

# Block ICMP (Ping)
iptables -A INPUT -p icmp -j DROP

# Save rules
```

```
iptables-save > /etc/iptables/rules.v4
```

OUTPUT:

Firewall rules were successfully applied and verified using the iptables -L command.

RESULT:

Thus, various configurations of IPTABLES firewall were successfully performed in Linux to control network traffic and enhance system security.